ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 2: E-commerce Business Performance and Customer Behaviour Analysis

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 2
Project Title	E-commerce Business Performance and Customer Behaviour Analysis
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 2 (Thursday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 2 Submission Form

1. PROJECT BACKGROUND

Maven Fuzzy Factory is an e-commerce company that wants to understand how its website performs, how customers interact with its products, which marketing channels generate sales, and how it can improve revenue and customer conversion.

Management has provided historical website traffic, pageview, order, product and refund data. Your responsibility is to analyse the available information and develop solutions that will support better business decisions.

Overall project objective: Use data and technology to identify business performance gaps, understand customer behaviour and recommend ways to improve e-commerce performance.

Track 1: Data Analytics

Assignment title: E-commerce Sales Performance and Customer Behaviour Analysis

Business objective: Analyse the company's sales, website traffic, product performance and refunds to help management understand its business performance.

Tools required:

- Microsoft Excel
- SQL (MySQL, PostgreSQL or SQL Server)
- Microsoft Power BI

Datasets to use:

- website_sessions.csv
- website_pageviews.csv
- orders_mav.csv
- order_items.csv
- order_item_refunds.csv
- products.csv
- Data dictionary

Tasks to complete

Task 1: Data understanding and preparation

1. Import the datasets and examine their structures.
2. Use the data dictionary to understand the fields.
3. Identify missing values, duplicates and inconsistent data types.
4. Identify the primary and foreign keys.
5. Establish the relationships between the datasets.
6. Clean and prepare the relevant datasets using Excel or Power Query.

Task 2: SQL business analysis

Write SQL queries to answer the following questions:

1. What is the total number of unique orders?
2. What is the total revenue generated?
3. Which five products generate the highest revenue?
4. Which product categories or product types perform best, where available?
5. Which marketing sources generate the most website sessions?
6. Which device type generates the most website traffic?
7. What is the website session-to-order conversion rate?
8. What is the total number and value of refunds?
9. Which months recorded the highest and lowest revenue?
10. What is the average order value?

Task 3: Power BI dashboard development

Develop an interactive dashboard containing:

- Total Orders
- Total Revenue
- Total Refunds
- Average Order Value

- Website Sessions
- Conversion Rate
- Revenue by Month
- Top Performing Products
- Sales by Product
- Sessions by Marketing Channel
- Sessions by Device Type
- Refund Analysis

Include appropriate filters and slicers.

Task 4: Business insights

Provide at least five findings from your analysis and explain how management can use them to improve business performance.

Expected deliverables

1. Cleaned datasets or documented transformation steps.
2. SQL scripts.
3. Power BI dashboard (.pbix).
4. Analytical report (PDF).
5. Five to seven business insights and recommendations.
6. A 5–10-minute video demonstrating the dashboard.

Track 2: Data Science

Assignment title: Exploratory Data Analysis of E-commerce Sales and Customer Behaviour

Business objective: Investigate customer behaviour, sales patterns and business performance using statistical and exploratory data analysis techniques.

Tools required:

- Python
- Google Colab or Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Datasets to use:

- website_sessions.csv
- website_pageviews.csv
- orders_mav.csv
- order_items.csv
- order_item_refunds.csv
- products.csv

Tasks to complete

Task 1: Data exploration

1. Load the datasets into Python.
2. Examine the shape, columns and data types.
3. Identify missing values and duplicate records.
4. Investigate the relationships between the datasets.
5. Clean and prepare the data for analysis.

Task 2: Exploratory data analysis

Investigate the following:

1. How does revenue change over time?
2. Which products generate the highest sales?
3. What are the patterns in customer purchasing behaviour?
4. Which marketing channels attract the most website sessions?
5. How does website traffic differ between mobile and desktop users?
6. What proportion of website sessions result in orders?
7. Which products have the highest refund activity?
8. Are there noticeable differences between new and returning website sessions?

Task 3: Statistical analysis and visualisation

1. Calculate relevant descriptive statistics.
2. Examine relationships between website traffic and orders.
3. Compare sales performance across different periods.
4. Investigate the relationship between product sales and refunds.
5. Create at least six meaningful visualisations.

Task 4: Interpretation and recommendations

Explain the patterns discovered and identify opportunities for improving sales, customer engagement and marketing performance.

Expected deliverables

1. Jupyter Notebook (.ipynb).
2. Python source code.
3. At least six visualisations.
4. EDA report (PDF).
5. Five to seven evidence-based business insights.
6. A 5–10-minute video presentation.

Track 3: Machine Learning Engineering

Assignment title: E-commerce Customer Purchase Conversion Prediction

Business objective: Maven Fuzzy Factory wants to identify website sessions that are more likely to result in purchases. The company wants to use this information to improve its marketing strategies and customer conversion rates.

Tools required:

- Python
- Google Colab or Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn
- Matplotlib and Seaborn

Datasets to use:

- `website_sessions.csv`
- `orders_mav.csv`
- `website_pageviews.csv` (optional for additional feature engineering)

Tasks to complete

Task 1: Understand the business problem

1. Explain what customer conversion means.
2. Identify the business problem the company wants to solve.
3. Define the machine learning objective.
4. Identify the target variable and potential input features.

Task 2: Data preparation

1. Load the website sessions and orders datasets into Python.
2. Merge the datasets using `website_session_id`.
3. Create a target variable called `converted`:
    - 1 = The website session resulted in an order.
    - 0 = The website session did not result in an order.
4. Identify and handle missing values.
5. Select appropriate features for modelling, such as:
    - Device type.
    - Marketing source.
    - Marketing campaign.
    - Marketing content.
    - Referral source.
    - Repeat session indicator.
6. Encode categorical variables.
7. Prepare the modelling dataset.

Task 3: Model development

1. Split the dataset into training and testing sets, preferably using a time-based split.
2. Develop a Logistic Regression classification model.
3. Train the model using the training dataset.
4. Generate predictions using the test dataset.
5. Compare actual and predicted conversion outcomes.

Task 4: Model evaluation

Evaluate the model using:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- Confusion matrix.

Explain what each evaluation metric tells the business.

Task 5: Business interpretation

1. Identify which features appear most useful for predicting conversion.
2. Explain how the business can use the results.
3. Identify limitations of the model.
4. Recommend improvements for future model development.

Expected deliverables

1. Prepared modelling dataset.
2. Jupyter Notebook (.ipynb).
3. Python source code.
4. Trained Logistic Regression model.
5. Model evaluation report.
6. Business recommendations.
7. A 5–10-minute video demonstration.

Important technical instruction: Do not use information generated after a session's purchase as a predictive feature. This would cause data leakage. The model should only use information available at the time the prediction is made.

Track 4: Generative AI Engineering

Assignment title: Development of an AI-Powered E-commerce Business Assistant

Business objective: Maven Fuzzy Factory wants an AI-powered assistant that can help its management team obtain quick answers to business questions relating to products, orders, sales, marketing performance and refunds.

Tools required:

- Python
- Google Colab or VS Code
- Streamlit
- OpenAI API or another accessible LLM provider
- Pandas
- Prompt engineering techniques

Datasets to use:

- orders_mav.csv
- order_items.csv
- products.csv
- order_item_refunds.csv
- website_sessions.csv
- maven_fuzzy_factory_data_dictionary.csv

Tasks to complete

Task 1: Business understanding

1. Review the Maven Fuzzy Factory business scenario.
2. Examine the supplied datasets and data dictionary.
3. Identify the types of business questions management may want answered.
4. Identify at least ten questions an e-commerce manager could ask an AI assistant.

Examples:

- What is the total number of orders?
- Which product generated the highest revenue?
- Which marketing channel attracted the most website sessions?
- What is the total refund amount?
- Which device type generated the most traffic?

Task 2: Prompt engineering

1. Design a system prompt that defines the assistant's role.
2. Create prompts for answering sales and product-related questions.
3. Create prompts for marketing and customer behaviour questions.
4. Create prompts for refund-related questions.

5. Test how changes in prompt instructions affect response quality.

Task 3: Build the AI assistant

1. Prepare relevant business summaries from the datasets using Python.
2. Connect the prepared information to an LLM through an API.
3. Build a simple chatbot interface using Streamlit.
4. Enable users to ask questions about the company's business performance.
5. Ensure the assistant uses the supplied business information when answering.
6. Display clear responses and explain when the available data cannot answer a question.

Task 4: Testing and evaluation

1. Test the assistant with at least ten business questions.
2. Compare its responses with verified calculations from the datasets.
3. Identify inaccurate or unsupported responses.
4. Document how the assistant handles questions outside its knowledge.
5. Recommend improvements.

Expected deliverables

1. Working AI assistant prototype.
2. Python source code.
3. Streamlit application.
4. System prompt and prompt documentation.
5. Ten test questions with expected and actual responses.
6. AI testing and evaluation report.
7. A 5–10-minute demonstration video.

Important: Interns should not upload confidential information or expose API keys in their code repositories. A basic data-grounded chatbot is sufficient for this week. Advanced RAG and vector databases can be introduced in subsequent weeks.

Track 5: Project Management

Assignment title: E-commerce Business Analysis Project Planning and Coordination

Business objective: Maven Fuzzy Factory has initiated a business performance improvement project. The management team needs a structured project plan to coordinate the analysis, technical development and delivery of business recommendations.

Tools required:

- Microsoft Word
- Microsoft Excel
- Trello or Jira
- Microsoft PowerPoint

- Microsoft Project (optional)

Resources to use:

- All seven supplied datasets.
- Maven Fuzzy Factory data dictionary.
- Week 2 business scenario.

Tasks to complete

Task 1: Project initiation

1. Review the business scenario.
2. Identify the business problem.
3. Define the project objectives.
4. Identify key stakeholders.
5. Define the expected business outcomes.

Task 2: Requirements gathering

Develop a Business Requirements Document (BRD) that captures:

1. Business background.
2. Business problem.
3. Project objectives.
4. Functional requirements.
5. Non-functional requirements.
6. Expected deliverables.
7. Project assumptions and constraints.

Task 3: Project planning

1. Develop a project charter.
2. Define the project scope and exclusions.
3. Create a Work Breakdown Structure (WBS).
4. Identify tasks and activities.
5. Assign responsibilities to the different project roles.
6. Develop a project timeline.
7. Identify project milestones.

Task 4: Project execution planning

1. Create a Trello or Jira project board.
2. Create task cards for the five internship tracks.
3. Organise the tasks into:
    - To Do
    - In Progress
    - Under Review
    - Completed

4. Define the project communication process.
5. Establish how progress will be monitored.

Task 5: Risk management

Identify at least five potential project risks.

Examples:

- Missing or inconsistent data.
- Delayed submission of technical deliverables.
- Incorrect interpretation of business requirements.
- Integration difficulties between different technical outputs.
- Poor communication between project teams.

For each risk, document its likelihood, potential impact and mitigation strategy.

Expected deliverables

1. Project Charter.
2. Business Requirements Document.
3. Work Breakdown Structure.
4. Project timeline.
5. Project Risk Register.
6. Trello/Jira project board.
7. Weekly Project Status Report.
8. Project presentation.

6. Common assessment rubric for Week 2

Assessment criteria	Weight
Understanding of business problem	10%
Technical implementation	30%
Problem-solving and analytical thinking	20%
Documentation	15%
Business insights and recommendations	10%
Presentation and communication	10%
Professionalism and submission	5%
Total	100%

Professional Development Requirement

As part of building your professional portfolio:

• Update your GitHub repository with your Week 2 project.

• Publish a LinkedIn post describing your project

• Publish an X (Twitter) post summarizing your project.

• Tag AnalystLab Africa on LinkedIn.

• Tag @analystlabafric on X.

• Include the hashtag #AnalystLabAfrica on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.